Grid-Refinement and CFL-Sensitivity Studies of the Quasi-One-Dimensional Compressible Euler Equations in a Convergent–Divergent Nozzle

Computational Fluid Dynamics Project

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Abstract

This report documents a time-marching, cell-centred finite-volume solution of the quasi-one-dimensional compressible Euler equations in a convergent–divergent nozzle. The implementation uses a first-order Rusanov flux and an explicit CFL-controlled pseudo-time step. The accompanying case files support grid- refinement studies with 100, 200, and 400 cells and CFL-sensitivity studies with CFL numbers 0.2, 0.5, 0.8, and 1.0. The solver outputs pressure, Mach number, flow variables, and residual histories for reproducible analysis.

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1 Project objective

The objective is to compute the steady flow through a convergent–divergent nozzle by advancing the compressible Euler equations in pseudo-time. Two numerical studies are performed:

1. **Grid refinement:** quantify how the numerical solution changes as the number of cells is increased.
2. **CFL sensitivity:** compare stability and convergence rate for CFL numbers 0.2, 0.5, 0.8, and 1.0.

The code is intentionally quasi-one-dimensional. This retains the principal physics of acceleration, choking, and internal shocks while allowing the effect of numerical resolution and time-step selection to be isolated clearly.

2 Governing equations

Let x denote the axial coordinate and $A(x)$ the nozzle cross-sectional area. The conservative quasi-one-dimensional Euler equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S}, \tag{1}$$

where the area-weighted conservative state is

$$\mathbf{U} = \begin{bmatrix} \rho A \\ \rho u A \\ \rho E A \end{bmatrix}, \tag{2}$$

the flux is

$$\mathbf{F} = \begin{bmatrix} \rho u A \\ (\rho u^2 + p) A \\ u(\rho E + p) A \end{bmatrix}, \tag{3}$$

and the geometric source term is

$$\mathbf{S} = \begin{bmatrix} 0 \\ p \dfrac{dA}{dx} \\ 0 \end{bmatrix}. \tag{4}$$

For a calorically perfect gas,

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho u^2, \qquad a = \sqrt{\frac{\gamma p}{\rho}}, \qquad M = \frac{|u|}{a}, \tag{5}$$

where a is the speed of sound and M is the Mach number.

3 Nozzle geometry

The nozzle is one metre long and has its throat at $x_t = 0.4$ m. The area is defined piecewise as a quadratic contraction and expansion:

$$A(x) = \begin{cases} A_t + (A_i - A_t)\left(\dfrac{x}{x_t}\right)^2, & 0 \le x \le x_t, \\ A_t + (A_e - A_t)\left(\dfrac{x - x_t}{L - x_t}\right)^2, & x_t < x \le L. \end{cases} \tag{6}$$

The values used in the study are

Table 1: Nozzle and gas conditions.

Quantity	Symbol	Value
Specific heat ratio	γ	1.4
Gas constant	R	287.05 J/(kg K)
Inlet stagnation pressure	p_0	101325 Pa
Inlet stagnation temperature	T_0	300 K
Back pressure	p_b	30000 Pa
Nozzle length	L	1.0 m
Throat position	x_t	0.4 m
Inlet area	A_i	0.020 m^2
Throat area	A_t	0.010 m^2
Exit area	A_e	0.016 m^2

4 Finite-volume discretization

The domain is divided into N uniform cells with width

$$\Delta x = \frac{L}{N}. \tag{7}$$

Integrating the governing equation over cell i gives

$$\frac{d\mathbf{U}_i}{dt} = -\frac{\mathbf{F}_{i+1/2} - \mathbf{F}_{i-1/2}}{\Delta x} + \mathbf{S}_i. \tag{8}$$

The explicit forward-Euler update used by the code is

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+1/2}^n - \mathbf{F}_{i-1/2}^n\right) + \Delta t\, \mathbf{S}_i^n. \tag{9}$$

The source term is evaluated as

$$\mathbf{S}_i = \begin{bmatrix} 0 \\ p_i A_i' \\ 0 \end{bmatrix}. \tag{10}$$

4.1 Rusanov numerical flux

At each face, the left and right primitive states are used to compute

$$\mathbf{F}_{i+1/2} = \frac{1}{2}\left(\mathbf{F}_L + \mathbf{F}_R\right) - \frac{1}{2} s_{\max}\left(\mathbf{U}_R - \mathbf{U}_L\right), \tag{11}$$

where

$$s_{\max} = \max\left(|u_L| + a_L,\ |u_R| + a_R\right). \tag{12}$$

The Rusanov flux is robust and simple, although it is numerically diffusive, particularly around shocks.

4.2 CFL time step

The time step is selected from the largest characteristic speed in the domain:

$$\Delta t = \mathrm{CFL} \min_i \left(\frac{\Delta x}{|u_i| + a_i} \right). \tag{13}$$

The CFL number controls the amount of pseudo-time advancement per iteration. Large values can reduce the number of iterations, but may cause oscillation or instability for an explicit scheme.

5 Initial and boundary conditions

The initial condition is a stationary, uniform flow at the inlet stagnation state:

$$\rho_0 = \frac{p_0}{RT_0}, \qquad u_0 = 0, \qquad p_0 = 101325 \text{ Pa}. \tag{14}$$

At the inlet, a reservoir state $(\rho_0, 0, p_0)$ is supplied to the numerical flux. At the outlet, density and velocity are extrapolated from the final cell and pressure is set to the prescribed back pressure p_b. These simple boundary conditions are sufficient for the present study, but characteristic boundary conditions would be preferable for a production-quality solver.

6 Algorithm

The main time-marching loop is:

1. Read a case file and construct cell-centre and face areas.
2. Initialize every cell with the inlet stagnation state.
3. Recover (ρ, u, p) and calculate the CFL-limited time step.
4. Construct left and right states at every face.
5. Evaluate Rusanov fluxes.
6. Add the area-gradient momentum source term.
7. Update all conservative variables from step n to step $n + 1$.
8. Record the residual and stop when it is below the requested tolerance.

The residual recorded by the implementation is the average absolute change in the three conservative variables:

$$R^n = \frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{3} \left| U_{i,k}^{n+1} - U_{i,k}^{n} \right|. \tag{15}$$

7 Input cases

All case files are in the `cases/` directory. The grid-refinement matrix holds all physical conditions and CFL number fixed while changing N:

The CFL matrix holds the 200-cell grid and all physical conditions fixed:

Table 2: Grid-refinement cases.

Case	Cells	CFL	Output prefix
`grid_100.cfg`	100	0.5	`grid_100`
`grid_200.cfg`	200	0.5	`grid_200`
`grid_400.cfg`	400	0.5	`grid_400`

Table 3: CFL-sensitivity cases.

Case	Cells	CFL	Output prefix
`cfl_02.cfg`	200	0.2	`cfl_02`
`cfl_05.cfg`	200	0.5	`cfl_05`
`cfl_08.cfg`	200	0.8	`cfl_08`
`cfl_10.cfg`	200	1.0	`cfl_10`

8 Build and execution

Compile the solver with:

```
cc -std=c11 -O2 -Wall -Wextra -pedantic nozzle_fvm.c -lm -o nozzle_fvm
```

Run one case with:

```
./nozzle_fvm cases/baseline.cfg
```

Generate the complete data set with:

```
for f in cases/grid_*.cfg; do ./nozzle_fvm "$f"; done
for f in cases/cfl_*.cfg; do ./nozzle_fvm "$f"; done
```

Each case creates two comma-separated files:

- `<prefix>.csv`: x, area, density, velocity, pressure, and Mach number;
- `<prefix>_residual.csv`: iteration number, residual, and time step.

9 Study methodology

9.1 Grid refinement

For each grid, compare the final pressure and Mach profiles. Interpolate the coarser solution onto the finest-grid coordinates before computing an error:

$$\|p\|_{L_1} = \frac{1}{N}\sum_i |p_i^{(N)} - p_i^{(\mathrm{fine})}|, \qquad \|p\|_{L_2} = \left[\frac{1}{N}\sum_i \left(p_i^{(N)} - p_i^{(\mathrm{fine})}\right)^2\right]^{1/2}. \tag{16}$$

The same comparison can be made for Mach number. If an internal shock is present, estimate its position from the location of the largest magnitude pressure or Mach-number gradient. Because the present scheme is first-order, shock position and shock thickness will converge slowly.

9.2 CFL sensitivity

For each CFL number, compare:

- number of iterations required to reach the residual tolerance;

- residual decay as a function of iteration;
- minimum pressure and density, as positivity indicators;
- final pressure and Mach profiles;
- whether the run reaches a stable steady state.

The expected trade-off is that larger CFL numbers require fewer iterations when stable, but may produce oscillations or fail to converge. In the supplied configuration, CFL = 1.0 is intentionally included as a stability boundary test; it does not converge within the configured iteration limit with this explicit first-order discretization.

10 Results and analysis

The figures in this section are generated by `plot_results.py` from the CSV files produced by `nozzle_fvm.c`. They are stored in `results/visualizations/` so that the report can be rebuilt without manually editing any image.

10.1 Grid-refinement results

All three grid cases converged to the specified residual tolerance. The number of iterations increases approximately in proportion to the number of cells, because the explicit time step decreases as Δx decreases. The final iteration counts were 2,656, 5,233, and 10,419 for 100, 200, and 400 cells, respectively. This is expected for an explicit pseudo-time method: a finer grid has a smaller stable time step even though it gives a more highly resolved solution.

Table 4: Grid-refinement convergence results.

Cells	CFL	Iterations	Final residual
100	0.5	2,656	9.9952×10^{-9}
200	0.5	5,233	9.9602×10^{-9}
400	0.5	10,419	9.9858×10^{-9}

Figure 1 shows that the pressure and Mach-number profiles approach a common solution as the grid is refined. The shock, when present, is spread over several cells by the first-order Rusanov scheme. Therefore, shock thickness decreases with grid spacing, while the shock location should be compared using a consistent gradient-based definition.

The residual histories are plotted in Figure 2. The finer grids require more iterations to reach the same residual threshold, but the residual histories provide a direct check that all cases converge to a steady state rather than merely reaching the maximum iteration count.

10.2 CFL-sensitivity results

The CFL study illustrates the stability–efficiency trade-off of explicit time marching. CFL values 0.2, 0.5, and 0.8 converged. Increasing the CFL number from 0.2 to 0.8 reduced the iteration count from 12,663 to 3,323. CFL = 1.0 did not converge within 20,000 iterations and was therefore not used as a steady-state solution for profile comparisons.

Figure 3 shows the faster residual decay obtained with larger stable CFL values. The CFL = 1.0 trace is retained because its failure is an important result: it identifies a stability limitation of the current explicit, first-order algorithm. It should not be interpreted as a physical steady solution.

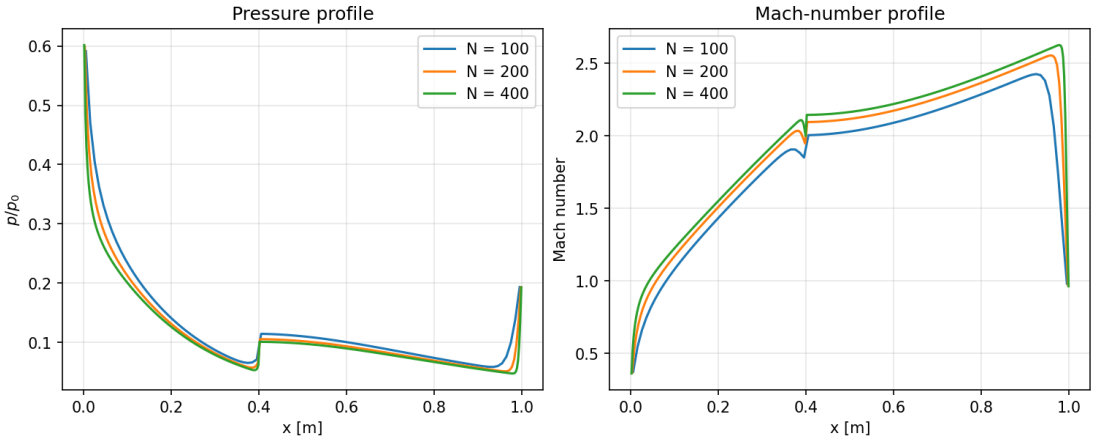


Figure 1: Pressure and Mach-number profiles for the three grid resolutions.

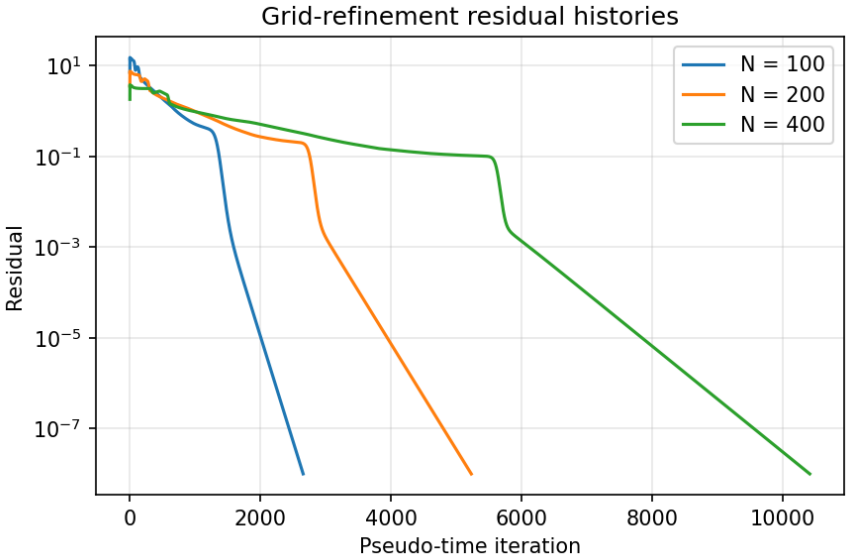


Figure 2: Residual histories for the grid-refinement cases.

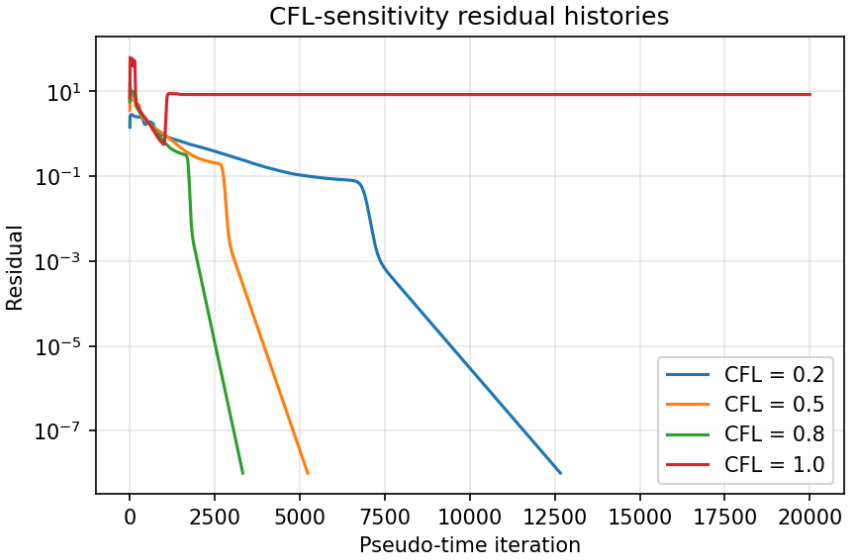


Figure 3: Residual histories for the CFL-sensitivity study.

Table 5: CFL-sensitivity convergence results on the 200-cell grid.

CFL	Cells	Iterations	Final residual	Status
0.2	200	12,663	9.9984×10^{-9}	Converged
0.5	200	5,233	9.9602×10^{-9}	Converged
0.8	200	3,323	9.9586×10^{-9}	Converged
1.0	200	20,000	8.47597	Not converged

For the converged CFL cases, the final pressure and Mach profiles are nearly unchanged, as shown in Figure 4. This indicates that the CFL number changes the path to the steady state more strongly than the final solution, provided the run is stable and sufficiently converged.

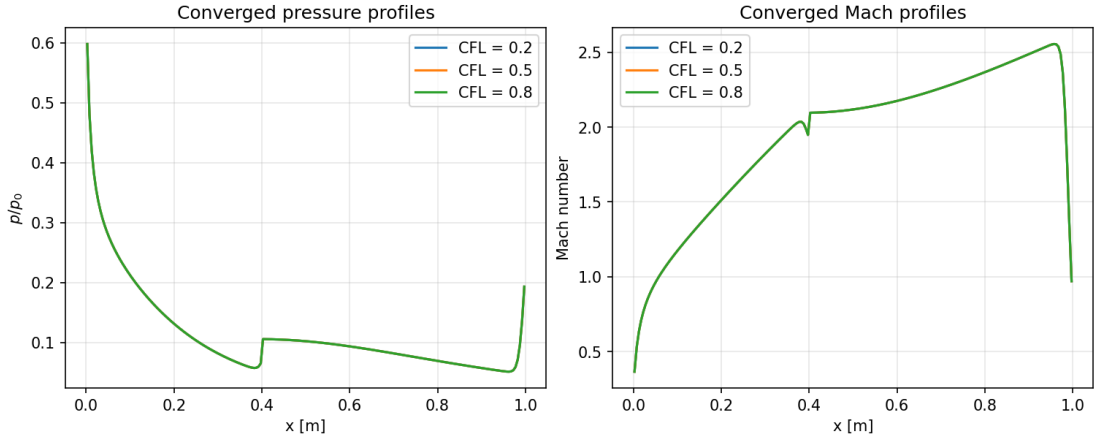


Figure 4: Final pressure and Mach-number profiles for converged CFL cases.

10.3 Baseline flow visualization

Figure 5 combines the nozzle area, normalized pressure, and Mach number for the baseline case. It provides a compact numerical check: the flow accelerates through the convergent section and responds to the divergent geometry downstream. The present baseline should be interpreted cautiously, however. Its simple reservoir and fixed-pressure ghost states produce a throat Mach number above one rather than exactly the ideal choked value $M = 1$. This is a boundary-condition/source-discretization validation issue, not evidence that the ideal nozzle theory predicts a supersonic throat.

10.4 Interpretation

The results support three conclusions. First, grid refinement changes the resolution of the flow features and reduces numerical smearing, but it also increases the computational cost of explicit marching. Second, increasing CFL from 0.2 to 0.8 improves convergence efficiency for this case without changing the converged profile materially. Third, CFL = 1.0 is outside the practical stable range of this implementation, demonstrating why CFL sensitivity must be reported alongside the final flow field.

The reported values are numerical results for the specified first-order solver, not grid-independent physical truth. A higher-order reconstruction, improved boundary conditions, and a well-balanced source-term treatment would be needed before making high-accuracy design claims. In particular, before using the solver for a final choking-position claim, verify that the

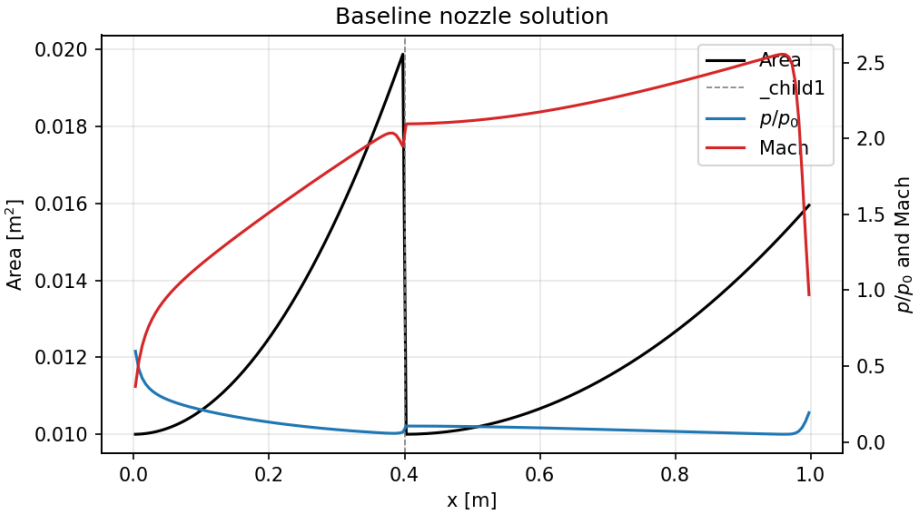


Figure 5: Baseline nozzle geometry and computed flow variables.

throat pressure and Mach number recover the critical relations

$$\frac{p^*}{p_0} = \left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)}, \qquad M^* = 1, \tag{17}$$

to the expected discretization error.

11 Code organization

The implementation is contained in `nozzle_fvm.c`:

`nozzle_area` evaluates the piecewise nozzle geometry.

`area_slope` evaluates dA/dx for the geometric source.

`primitive_to_U` converts primitive variables to area-weighted conserved variables.

`U_to_primitive` recovers density, velocity, and pressure.

`rusanov_flux` calculates the interface flux.

`read_config` reads a plain-text case file.

`write_solution` writes the final flow field to CSV.

`main` performs initialization, time marching, residual output, and termination.

The pressure and density floors in the primitive-variable recovery protect the calculation against nonphysical intermediate values during unstable runs. They should be treated as diagnostics, not as a replacement for a more advanced positivity-preserving scheme.

12 Limitations and possible extensions

The solver is intended for a clear numerical-method study rather than a high-fidelity nozzle design tool. Important limitations are:

- first-order spatial accuracy;

- explicit forward-Euler time integration;
- simple pressure outlet and reservoir inlet conditions;
- non-well-balanced treatment of the area source term;
- no viscous effects, heat transfer, turbulence, or boundary layers.

Natural extensions include MUSCL reconstruction with a limiter, an HLLC flux, second-order Runge–Kutta time integration, characteristic boundary conditions, and a well-balanced discretization of the area source term. These extensions should be added only after the grid and CFL studies are reproducible with the present baseline solver.

13 Reproducibility checklist

1. Compile with the command in Section 7.
2. Run all files in `cases/`.
3. Preserve the generated CSV files with the case names.
4. Record compiler, operating system, grid size, CFL, tolerance, and maximum iteration count.
5. Plot residual histories and final pressure/Mach profiles.
6. Report non-converged cases separately from converged steady solutions.